Experiment-17

17. DES Decryption Using Reverse Round Keys

Aim:

To write a C program to decrypt a DES ciphertext block using the round keys in reverse order.

Procedure:

1. Read the 64-bit DES key and ciphertext in hexadecimal.
2. Generate the DES round keys.
3. Apply the DES decryption operation using reverse round-key order.
4. Convert the decrypted block into hexadecimal.
5. Display the plaintext block.

Code (q17.c)

```
#include <stdio.h>

#include <openssl/des.h>


int main(void) {

  DES_cblock key, in, out;

  DES_key_schedule ks;

  char k[17], c[17];

  unsigned int x;

  int i;


  printf("Enter 16 hex digits for key: ");

  scanf("%16s", k);

  printf("Enter 16 hex digits for ciphertext: ");

  scanf("%16s", c);
```

```
    for (i = 0; i < 8; i++) {
        if (sscanf(k + 2*i, "%2x", &x) != 1) return 1;
        key[i] = (unsigned char)x;
        if (sscanf(c + 2*i, "%2x", &x) != 1) return 1;
        in[i] = (unsigned char)x;
    }

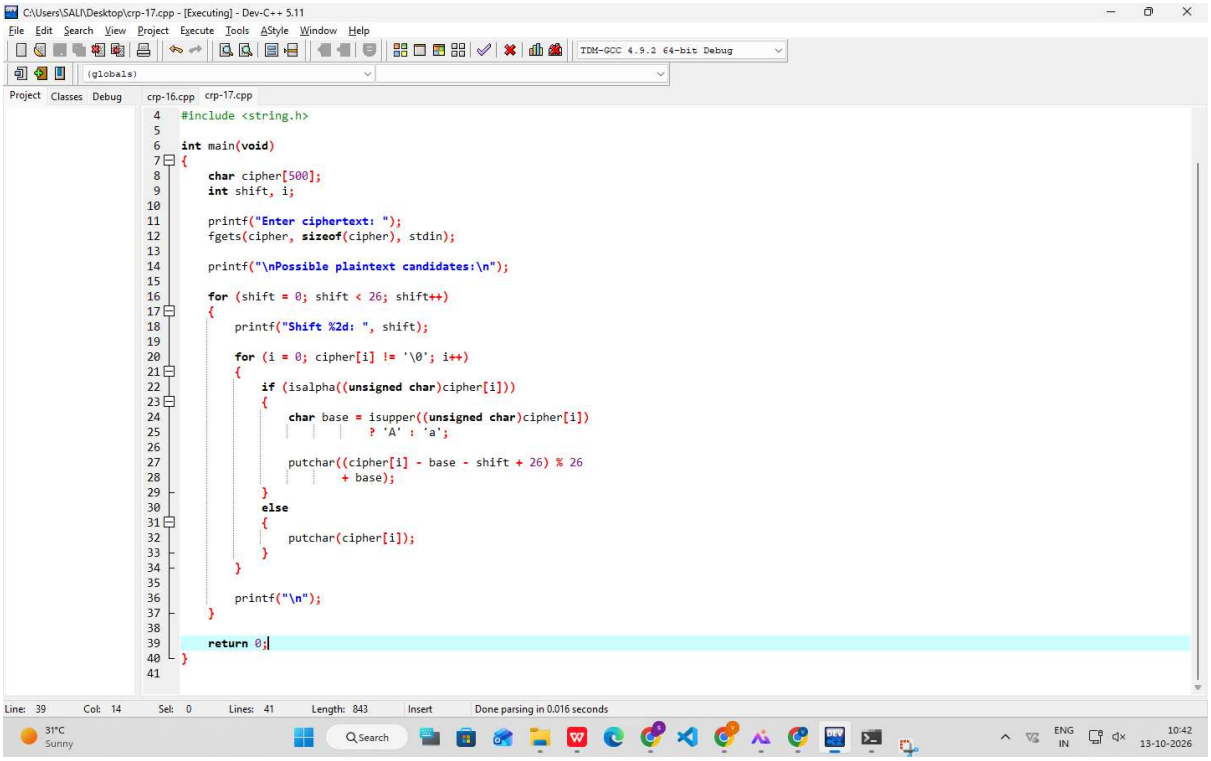
    DES_set_key_unchecked(&key, &ks);
    DES_ecb_encrypt(&in, &out, &ks, DES_DECRYPT);

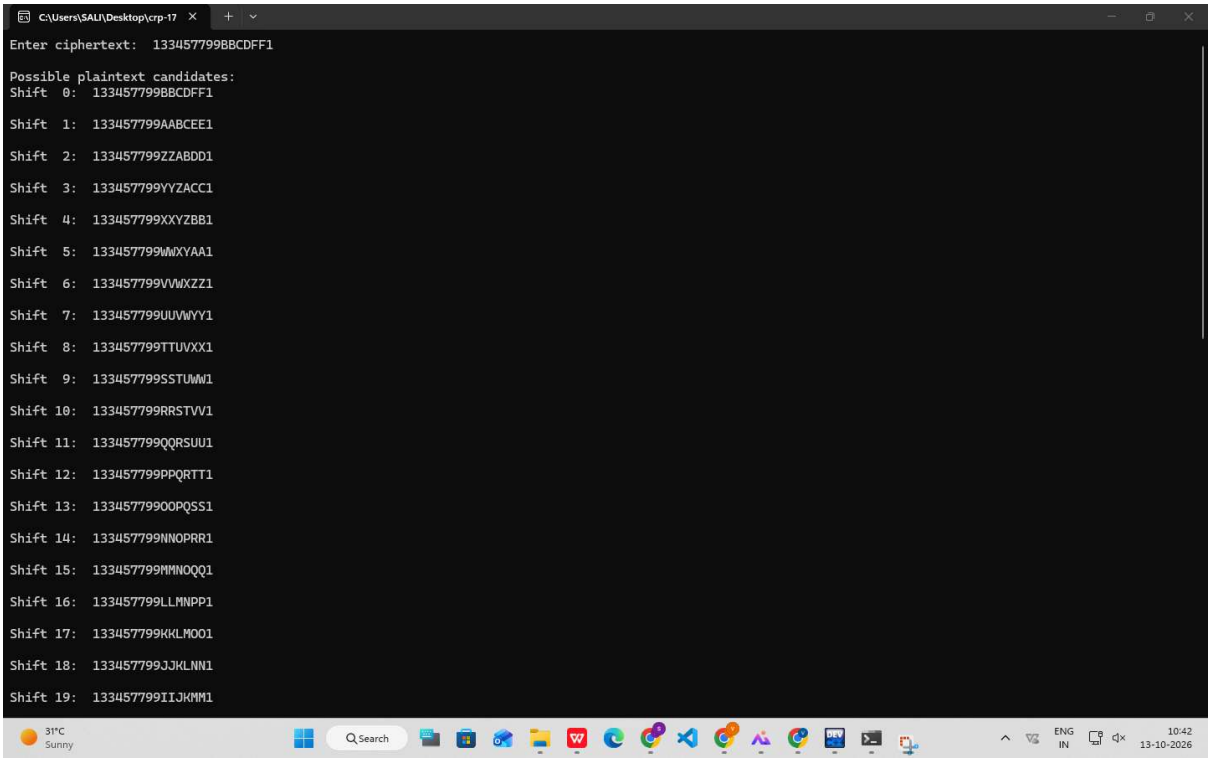
    printf("Decrypted plaintext: ");
    for (i = 0; i < 8; i++) printf("%02X", out[i]);
    printf("\n");
    return 0;
}
```

Sample Input

```
Enter 16 hex digits for key: 133457799BBCDFF1
Enter 16 hex digits for ciphertext: 85E813540F0AB405
```

Sample Output:





Result:

The DES ciphertext is decrypted successfully using reverse round-key order.